

Jehan Hasan

Undergraduate Teaching Assistant, CSE Department, United International University, Dhaka, Bangladesh

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RESEARCH INTERESTS

- **Natural Language Processing:** Retrieval-Augmented Generation (RAG) and Agentic AI Applications
- **Green Computing:** Optimization of Long-Term and Short-Term Memory Systems for Energy-Efficient Computing
- **Bioinformatics:** Computational Analysis of Drug–Chemical–Gene Interactions

EDUCATION

United International University, Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering

Feb 2022 — Jun 2026
CGPA: **3.92**/4.00

Major: Data Science

Thesis: Personalized Nutrition Management AI Agent Using LLM

Elective Courses: Machine Learning, Digital Image Processing, Bioinformatics, Compiler Design

St. Gregory's High School and College, Dhaka, Bangladesh
Higher Secondary Certificate

2017 — 2019
GPA: 4.25/5.00

Dhaka Collegiate School, Dhaka, Bangladesh
Secondary School Certificate

2009 — 2017
GPA: 5.00/5.00

ACADEMIC EXPERIENCE

United International University
Undergraduate Teaching Assistant

Dhaka, Bangladesh
Jun 2024 – Jun 2026

- Assisted in teaching undergraduate laboratory and theory courses across three consecutive semesters:
 - **Summer 2024:** Introduction to Computer Systems
 - **Fall 2024:** Data Structure and Algorithms II Laboratory
 - **Fall 2025:** Data Structure and Algorithms I Laboratory
- Conducted weekly laboratory sessions for approximately 40 undergraduate students, graded assignments and examinations, and provided one-on-one academic support to students.
- Developed course materials, prepared lab exercises, and conducted review sessions to enhance student learning.

RESEARCH EXPERIENCE

AdaptiveBeliefMem

[GitHub]

Memory-Augmented LLM Research

- Developed a memory framework that tightly couples an implicit belief state with an uncertainty-gated explicit memory mechanism, mediated by a frozen reasoning model.
- Learns when to externalize structured memory, enabling bounded, interpretable inference while handling sparse evidence across arbitrarily long conversation trajectories.

AgenticNutrition

[GitHub]

Agentic AI Development

- Engineered a context-aware AI agent for personalized nutrition management using LangGraph and LangChain.
- Leveraged federated data, EHR-based RAG, and real-time factors to generate disease-specific diet plans via an interactive chatbot.

FedDDoS

[GitHub]

Machine Learning based Federated Learning Development

- Developed a federated learning-based DDoS attack detection system for distributed networks.
- Enabled privacy-preserving collaborative threat detection without sharing raw data.

SOFTWARE PROJECTS

Melodia Academy

[GitHub]

Full-Stack Developer

- Developed a web-based Music School Enrollment & Lesson Scheduling System using Laravel, SQLite, and TailwindCSS.
- Managed instructors, courses, student enrollment, scheduling, learning materials, communication, and performance tracking.

NutriPick

[GitHub]

Full-Stack Development

- Developed an AI-driven personalized nutrition and fitness platform with diet recommendations, calorie prediction, meal planning, and chatbot support.
- Built using Django, SQLite, HTML, and TailwindCSS.

BuildIt

[GitHub]

Full-Stack Development

- Built a full-stack real estate platform for managing property listings, building projects, sales transactions, and contractor workflows.
- Implemented using Django, SQLite, HTML, and TailwindCSS.

Calculus MCP

[GitHub]

Agentic AI Development

- Created a production-ready MCP server bringing symbolic calculus capabilities to Claude Desktop.
- Implemented differentiation, integration, limit evaluation, and ODE solving using FastMCP, SymPy, and Python.

AWARDS

Academic Scholarship - United International University

Dhaka, Bangladesh

- **100% Scholarship:** 4 trimesters (top 2% student)
- **50% Scholarship:** 4 trimesters (top 6% student)
- **25% Scholarship:** 2 trimesters (top 10% student)

Rank 390 - Global Mathematical Talent Probe

Dhaka, Bangladesh

Grade 11

2017

SKILLS

- **Programming Languages:** C, C++, Python, Java, PHP
- **Data Science Libraries:** LangChain, LangGraph, PyTorch, TensorFlow, FastMCP, Scikit-learn, Pandas, NumPy, Matplotlib
- **Web Frameworks:** Django, Laravel, Flask
- **Database:** SQLite3, MySQL Server
- **Tools & Platforms:** Git, GitHub, Docker, Linux
- **Project Management:** Jira

REFERENCES

Dr. Jannatun Noor Mukta

Associate Professor & Director, Data Science Program, Department of Computer Science and Engineering, United International University, Dhaka, Bangladesh

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Dr. Mohammad Nurul Huda

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